EVERY LOCALLY FINITE TYPE MORPHISM INTO A REGULAR BASE SCHEME IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17801-17806 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17808-17811, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One-line 'See' pointer: kept because the statement is fully self-contained and the citation is exact to the label; flagged here for consistency with the BORDERLINE tally.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma more-algebra-lemma-regular-perfect-ring-map.

Source reference. Original lemma 067K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17811–17811. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume S is regular and f is locally of finite type. Then f is perfect.*

Proof. See More on Algebra, Lemma 067K. □